

Advanced (Habanero)

- Q1.** What is the solution to the system $x + y = 2$ and $x - y = 0$?
(A) $(x, y) = (1, 1)$
(B) $(x, y) = (0, 0)$
(C) $(x, y) = (2, 0)$
(D) $(x, y) = (0, 2)$
(E) No solution
- Q2.** What is the graph of $y = 2x - 1$ and $y = -x + 3$?
(A) Two parallel lines
(B) Two intersecting lines
(C) One line
(D) No lines
(E) A curve
- Q3.** Solve $x + 2y = 4$ and $3x - 2y = 2$ graphically.
(A) $(2, 1)$
(B) $(1, 2)$
(C) $(0, 0)$
(D) $(4, 0)$
(E) $(0, 4)$
- Q4.** Find the solution set of $y = x$ and $y = -x + 2$.
(A) $(1, 1)$
(B) $(0, 0)$
(C) $(1, 1)$ and $(-1, -1)$
(D) No solution
(E) $(0, 2)$
- Q5.** Graph $2x - y = 1$ and $x + 2y = 5$. What is the point of intersection?
(A) $(3, 1)$
(B) $(1, 2)$
(C) $(2, 3)$
(D) $(0, 0)$
(E) $(1, 1)$
- Q6.** What is the solution to $y = 2x + 1$ and $y = x - 2$?
(A) $(-1, -1)$
(B) $(1, 3)$
(C) $(-3, -5)$
(D) $(0, 0)$
(E) $(1, 1)$
- Q7.** Solve $y = x - 3$ and $y = -2x + 1$ graphically.
(A) $(2, -1)$
(B) $(-2, -5)$
(C) $(2, 1)$
(D) $(-1, -4)$
(E) $(0, 0)$
- Q8.** Graph $y = -x + 2$ and $y = x$. What is the intersection point?
(A) $(2, 0)$
(B) $(0, 2)$
(C) $(0, 0)$
(D) $(1, 1)$
(E) $(-1, 1)$

Open Ended Questions (FRQ)

- Q9.** Solve the system $x + 2y = 3$ and $2x - y = 2$ graphically and find the point of intersection. Show your work.
- Q10.** Graph the system $y = x - 1$ and $y = -x + 3$. Find the solution set and describe the relationship between the lines.

Chef's Tips

- Tip 1: Remind students to label axes and include titles on their graphs.
- Tip 2: Emphasize the importance of precise calculations when finding intersection points.
- Tip 3: Encourage students to verify their graphical solutions algebraically when possible.